

JAKE WALKLATE

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Summary

Multi-disciplinary engineer with 10 years of work experience in delivering production systems in early-stage startups and independent R&D environments and customer focused experiences. Strong background in programming, applied machine learning, and developer tooling. Proven ability to take products from concept to launch. Seeking a stable role with long-term sustainable growth and diverse responsibilities with the flexibility to support PhD research.

Education

Bachelor of Science & Engineering (MedSci & Robotics), from UTS (University of Technology Sydney)	<i>Expected 2027</i>
Cert IV in IT (Cyber Security & Software Development), from TAFE NSW	<i>Graduated 2020</i>

Skills Overview

Frontend & Mobile React (+Next.JS), Flutter, Tailwind, Angular, Bootstrap	Healthcare Cell Culturing, Gene Recombination, iPSCs, Fiji, 3D Bioprinting, Data Privacy
Backend & Systems - Node.js, REST APIs - SQL (MySQL, PostgreSQL), basic NoSQL (MongoDB) - Rust (Performance-focused applications) - Java & C# (Business logic)	Cyber Security Network security basics, secure software practices
Cloud, Tooling & Testing - Git, CI/CD concepts/workflows, Docker - Microsoft Azure, Google Cloud Platform & Vercel - Unit and Integration Testing (Jest, JUnit concepts)	Mechatronics - Circuit analysis & Digital Circuit Design (FPGA) - C++ (STM32 & Arduino Embedded Systems) - Computer Aided Drafting (CAD)
	Artificial Intelligence Neural Networks, Decision Trees, Fuzzy Logic

Abridged Experience

Systems Architect, at Coreable	<i>1 year</i>
- Built and deployed a production-ready web servers — from early prototype through to launch — by collaborating closely with UI/UX designers and business stakeholders - Designed backend and frontend components using React (Typescript), GraphQL, MySQL, and Node.JS - Owned system architecture and deployment decisions, ensuring reliability at launch by performing code reviews and managing CI/CD pipelines	
Software Developer, at ClockOn	<i>9 months</i>
Worked as part of a small corporate team building web/mobile applications and solutions using technologies such as C# .NET, Visual Studio, JavaScript, HTML, CSS (Bootstrap studio) and third-party UI components	

Independent Technical Projects

- Created and published a Visual Studio Code extension for Arduino .ino files (400+ downloads & 5-star reviews)
- R&D project exploring genetic clinical support with Flutter and Material Design; archived due to funding constraints
- Trained a neural network classifier in TensorFlow Keras (Python) for medical adjacent applications (CNN & MLP)
- Designed a 3D printable IoT robot arm with calculated celestial Low Earth Object tracking for Arduino (C++)
- Implemented a Web Assembly (WASM) Spiking Neural Network as a psychology research-based learning project
- Re-created a CHIP-8 virtual machine in Rust for CPU architecture research, compatible with existing video games
- Launched a Wikipedia-like information aggregation platform for diverse global viewpoints using Next.JS & Supabase
- Built a custom community forum & gaming server with MongoDB & EJS (JavaScript) for online PC gamers
- Modernizing biological lab research with a management platform built with Tauri (Rust), TypeScript, mDNS & SQLite

Additional Information

- Available full time (40 hours/week) on negotiation
- Soft skills include cross-disciplinary communication, team collaboration, and empathetic mentorship
- Engineers Australia Associate Member and WH&S Whitecard holder, former Australian Computing Society member
- Responsible Service of Alcohol and Responsible Conduct of Gambling license holder

References available on request.